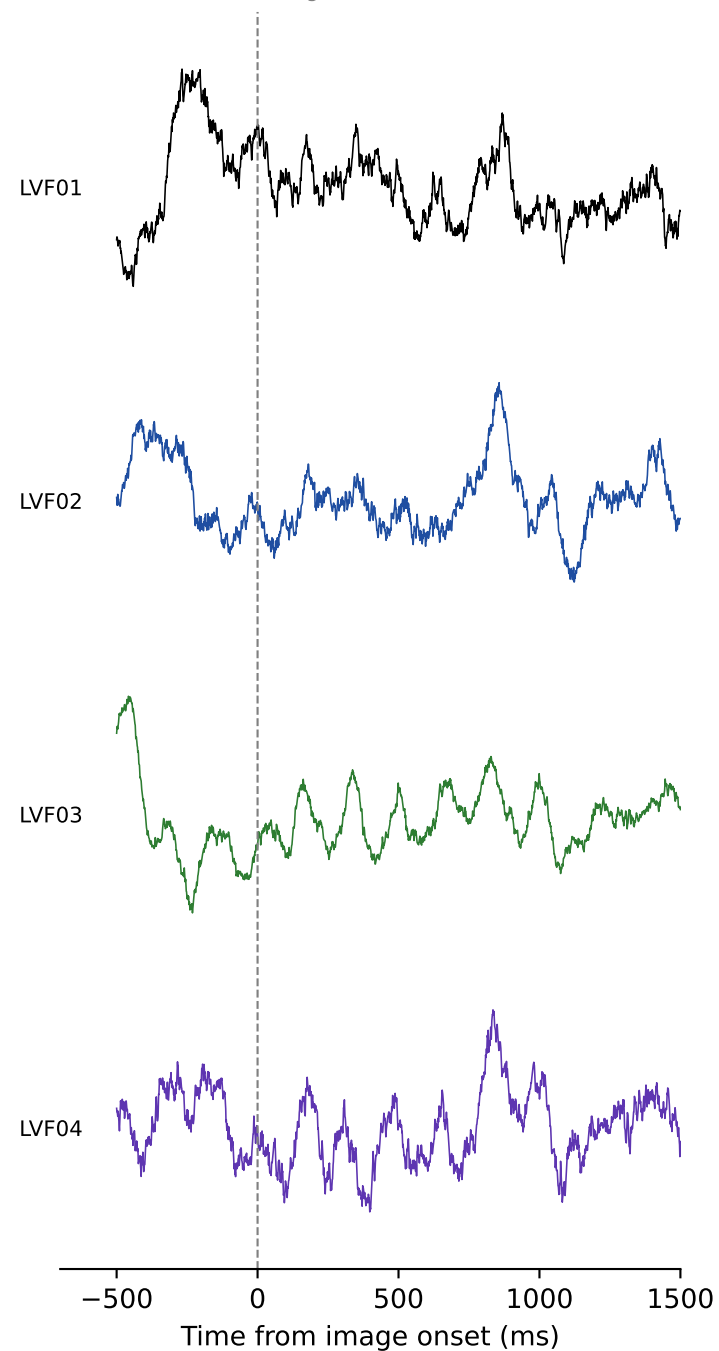


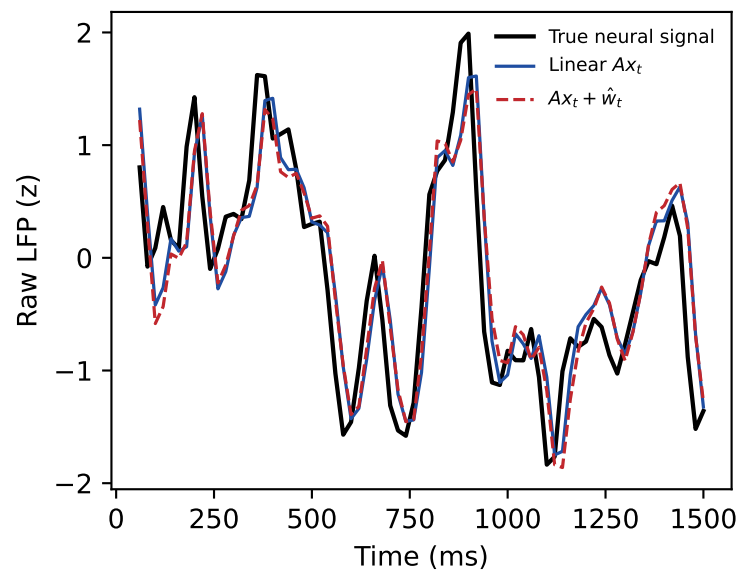
# Brain as locally linear system — RAW broadband LFP state: $x_{t+\Delta} = Ax_t + b + \hat{w}_t$ (P12 · LVF · MSIT)

## A Neural data & task context

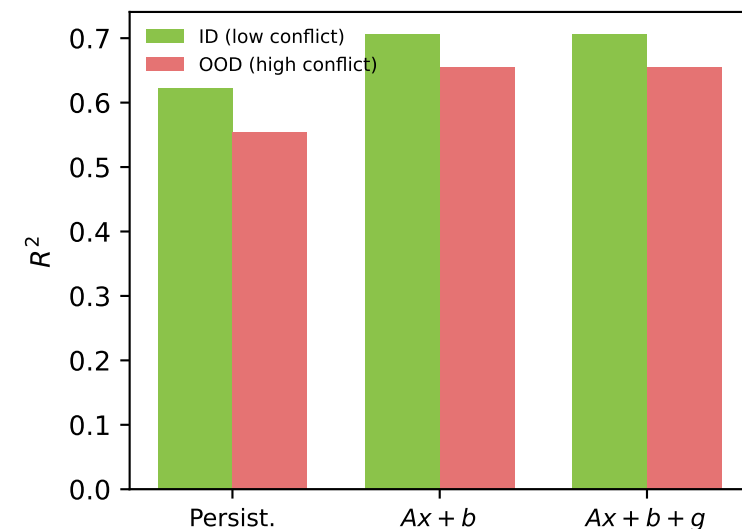
MSIT · P12 · LVF lead — RAW broadband LFP



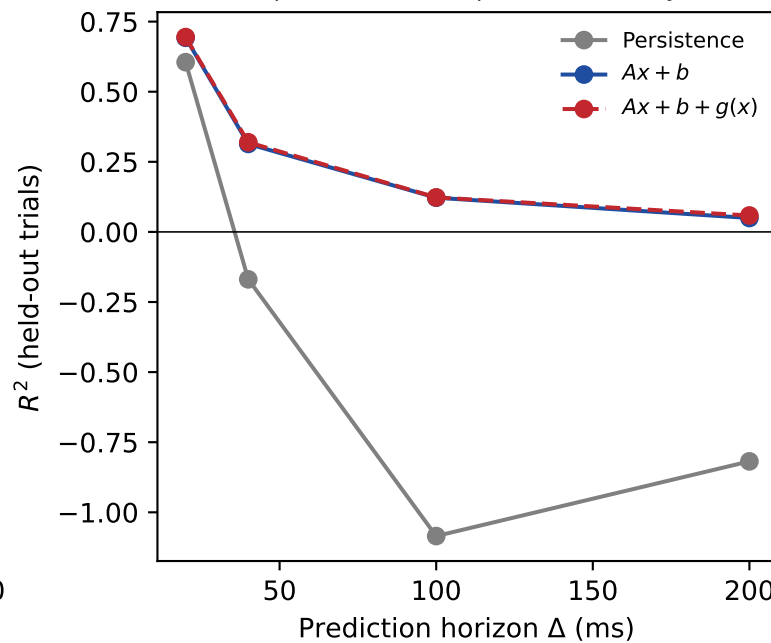
## B Linearization in-domain (raw LFP, $\Delta=20$ ms)



## C Context shift: low→high conflict frozen model, $\Delta=20$ ms



## Raw LFP: persistence collapses, $Ax + b$ stays informative



## Mild short-horizon OOD degradation ( $\Delta R^2 \approx -0.05$ at 20-40 ms)

